

Drugon: Phase-Encoded Morgan Fingerprints Match Pretrained Chemical Transformers in Low-Data Affinity Prediction

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Abstract

Pretrained chemical language models such as ChemBERTa and MoLFormer have become the default choice for molecular property prediction. However, they require large training sets to fine-tune effectively, and their computational footprint is substantial. We present **Drugon**, a compact alternative based on a *Phase-Encoded Morgan fingerprint* combined with a LightGBM regressor. Drugon requires approximately $15,000\times$ fewer parameters and $40\times$ less training time than fine-tuned transformers, while achieving *higher* predictive accuracy on three independent ChEMBL targets ($n = 1378$ to 2000). On the c-Abl dataset (ChEMBL4550), Drugon reaches a Pearson $r = 0.749$ under 5-fold cross-validation, compared with 0.605 for fine-tuned ChemBERTa and 0.551 for MoLFormer. Across all three targets, phase encoding provides a consistent, non-negative improvement over the classical binary Morgan fingerprint ($\Delta r \in [0.000, 0.026]$). Our results suggest that, in the low-data regime ($n < 2000$), fingerprint-based methods remain the most efficient and interpretable choice for target-specific affinity prediction.

Keywords: molecular fingerprints, binding affinity, low-data regime, gradient boosting, phase encoding.

1 Introduction

Predicting protein–ligand binding affinity is a central task in structure-based and ligand-based drug discovery. Accurate *in silico* estimation of binding free energy or its experimental proxies (pIC_{50} , $\text{p}K_i$, $\text{p}K_d$) can reduce the cost and time of hit identification by orders of magnitude [1, 2].

Over the past decade, the field has progressively shifted toward large-scale pretrained language models trained on SMILES strings. Early work on ChemBERTa [3] demonstrated that masked language modeling on 77 million molecules could produce useful chemical representations. This was followed by larger models such as MoLFormer [4], trained on 1.1 billion molecules, and by specialized architectures such as Uni-Mol [5] and Graphormer [6] that incorporate three-dimensional geometry.

These pretrained models achieve impressive results on MoleculeNet-style benchmarks [7], where training sets range from tens of thousands to hundreds of thousands of molecules. However, in many practical drug-discovery campaigns, the available training data is far smaller. A typical

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target-specific medicinal chemistry program might accumulate a few hundred to a few thousand active compounds before initiating lead optimization. In this *low-data regime*, the utility of pretrained transformers is significantly less clear.

In this work, we revisit a classical alternative: the Extended-Connectivity Fingerprint (ECFP, commonly known as the Morgan fingerprint) [8] combined with a gradient-boosted decision tree regressor [9]. We propose a compact extension, Drugon, in which each fingerprint bit is encoded as a *complex number* whose phase is derived from the local chemical environment of the contributing atoms. The real and imaginary components are concatenated into a real-valued feature vector that is passed to LightGBM.

Our contributions are:

1. We introduce **phase-encoded Morgan fingerprints**, a minimal extension that injects chemical context into a classical descriptor at negligible cost.
2. We show that Drugon *outperforms fine-tuned* ChemBERTa and MolFormer on the c-Abl target (ChEMBL4550), despite having $\sim 15,000\times$ fewer parameters and requiring $\sim 40\times$ less training time.
3. We evaluate Drugon on two additional, chemically distinct ChEMBL targets (AChE and COX-2), demonstrating that phase encoding provides a consistent, non-negative improvement over the binary Morgan baseline.
4. We argue that, in the low-data regime, *data scale rather than architecture* is the primary bottleneck for affinity prediction, and that compact, interpretable models should remain the default choice when $n \lesssim 2000$.

The remainder of this paper is organized as follows. Section 2 reviews related work on molecular fingerprints and pretrained chemical models. Section 3 describes the Drugon method. Section 4 details the experimental setup. Section 5 presents results on all three targets. Section 6 discusses implications and limitations. Section 7 concludes.

2 Related Work

2.1 Molecular Fingerprints

Molecular fingerprints have been the workhorse of ligand-based virtual screening for over two decades [10, 11]. The Extended-Connectivity Fingerprint (ECFP) introduced by Rogers and Hahn [8] remains one of the most widely used descriptors, and is available in essentially every cheminformatics toolkit, including RDKit [12].

ECFP encodes the presence (or count) of circular substructures up to a fixed radius. In its binary form, each bit is set to 1 if the corresponding substructure appears at least once in the molecule, and 0 otherwise. Several variants exist: count-based [8], weighted by atom properties [13], and continuous-valued [14]. Our work builds on this line by introducing *phase-encoded* bits, where the contribution of each matching atom is a unit-magnitude complex number rather than a binary or count value.

2.2 Pretrained Chemical Language Models

The success of transformer architectures [15] in natural language processing has motivated their application to molecular representation learning. ChemBERTa [3] applied RoBERTa-style masked language modeling to 77 million molecules from the ZINC database, and demonstrated transfer learning to MoleculeNet tasks. MolFormer [4] scaled this to 1.1 billion molecules and introduced linear attention to handle longer sequences. More recent models such as Uni-Mol [5],

Graphormer [6], and Equiformer [16] incorporate three-dimensional geometry through equivariant architectures.

These models achieve strong performance on well-populated benchmarks. However, several studies have questioned their advantage in data-scarce settings. [17] showed that a simple Morgan fingerprint with a random forest can match graph neural networks on standard benchmarks, and more recent work [18] found no statistically significant advantage of pretrained models over well-tuned fingerprints in scaffold-split evaluations. Our results are consistent with this literature and extend it by directly comparing against *fine-tuned* transformers in a target-specific, low-data regime.

2.3 Gradient-Boosted Trees for Molecular Property Prediction

Gradient-boosted decision trees (GBDTs), and in particular LightGBM [9], have repeatedly proven competitive on molecular property prediction tasks [19, 20]. Their ability to handle high-dimensional, sparse feature vectors and to provide feature importance estimates makes them a natural pairing with molecular fingerprints.

Our work follows this classical recipe and shows that, when paired with an appropriate fingerprint, GBDTs remain a strong baseline even in the era of pretrained transformers.

3 Method

3.1 Preliminaries: Morgan Fingerprints

Let a molecule be represented as a graph $G = (V, E)$ with atoms $v \in V$ and bonds $e \in E$. The Morgan fingerprint [8] with radius R and N bits is constructed by enumerating, for each atom v , all circular substructures of radius up to R , hashing them into an integer in $\{0, \dots, N - 1\}$, and recording their occurrence.

In the standard *binary* variant, the fingerprint is $\mathbf{b} \in \{0, 1\}^N$ with

$$b_i = \begin{cases} 1 & \text{if substructure } i \text{ occurs in } G, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

In the *count* variant, b_i is the number of occurrences.

3.2 Phase-Encoded Fingerprints

We generalize the Morgan fingerprint by allowing each bit to be a *complex number*. For each bit i , let $\mathcal{A}_i \subseteq V$ denote the set of atoms contributing to that bit. We define the complex bit as

$$z_i = \sum_{v \in \mathcal{A}_i} \exp(i\phi(v)), \quad (2)$$

where the phase $\phi(v)$ encodes the local chemical environment of atom v :

$$\phi(v) = \pi \cdot (\alpha_Z Z_v + \alpha_A \mathbb{K}[v \text{ aromatic}] + \alpha_q q_v), \quad (3)$$

with Z_v the atomic number, q_v the formal charge, and $\alpha_Z, \alpha_A, \alpha_q$ are scaling constants.

In all experiments we use

$$(\alpha_Z, \alpha_A, \alpha_q) = (0.15, 1.5, 1.2). \quad (4)$$

The final feature vector is the concatenation of the real and imaginary parts:

$$\mathbf{x} = [\Re(\mathbf{z}), \Im(\mathbf{z})] \in \mathbb{R}^{2N}. \quad (5)$$

Motivation. The phase $\phi(v)$ serves as a soft, continuous signature of the chemical context of an atom. Two chemically distinct substructures that hash to the same bit will accumulate different phases and thus produce different complex magnitudes $|z_i|$, allowing the downstream model to distinguish them.

The three terms in Equation (3) were chosen to reflect three orthogonal chemical axes:

- **Element type** (Z_v): distinguishes carbon, nitrogen, oxygen, sulfur, halogen, etc.
- **Aromaticity** ($\mathbb{K}[\cdot]$): separates aromatic from aliphatic environments, which differ substantially in π -stacking propensity.
- **Formal charge** (q_v): captures electrostatic contributions to binding.

3.3 Regression Model

The real feature vector $\mathbf{x} \in \mathbb{R}^{2N}$ is passed to a LightGBM regressor [9]. We use the following hyperparameters throughout:

$$\begin{aligned} &500 \text{ trees, } \eta = 0.03, \quad 31 \text{ leaves, } d_{\max} = 7, \\ &\text{min child} = 5, \quad \text{subsample} = 0.8, \quad \text{colsample} = 0.7. \end{aligned} \tag{6}$$

We deliberately refrain from extensive hyperparameter tuning: the goal of this paper is to compare fingerprint *representations* under a fixed, reasonable regressor.

3.4 Complexity

The phase-encoded fingerprint requires, in addition to the standard Morgan computation, one complex exponential and one complex addition per atom-bit pair. Since the total number of such pairs is bounded by $|V| \cdot R$ for each molecule, the additional cost is $O(|V| \cdot R)$ and is dominated by the cost of the underlying Morgan hashing.

The downstream regressor has approximately 3,000 trainable parameters (31 leaves $\times$ 100 trees $\times$ depth 7), compared with 46–47 million parameters for the pretrained transformers considered in our experiments.

4 Experiments

4.1 Datasets

We evaluate Drugon on three independent targets drawn from the ChEMBL database [21], chosen to span three distinct therapeutic areas and pocket topologies:

- **c-Abl** (ChEMBL4550): the BCR-ABL tyrosine kinase targeted by imatinib in chronic myeloid leukemia. Pocket is deep and narrow.
- **AChE** (ChEMBL220): acetylcholinesterase, a target for Alzheimer’s disease. Pocket is a long, narrow gorge.
- **COX-2** (ChEMBL230): cyclooxygenase-2, targeted by NSAIDs for inflammation. Pocket is hydrophobic and shallow.

For each target we downloaded all binding assays (`assay_type = 'B'`) with a reported pChEMBL value in the range [4.0, 11.0], canonicalized the SMILES, removed duplicates, and filtered to molecules with 15–50 heavy atoms. The resulting dataset sizes and label distributions are summarized in Table 1.

Table 1: Datasets used in this work.

Target	ChEMBL ID	n	pIC ₅₀ range
c-Abl	CHEMBL4550	1378	5.60 – 9.47
AChE	CHEMBL220	2000	4.00 – 10.93
COX-2	CHEMBL230	2000	4.00 – 10.70

4.2 Baselines

We compare Drugon against the following baselines:

1. **Binary Morgan** (radius 2, 1024 bits) with LightGBM. This is the standard classical baseline.
2. **ChemBERTa (fine-tuned)** [3], fine-tuned end-to-end for 15 epochs with AdamW (learning rate 2×10^{-5} , batch size 32).
3. **MoLFormer (fine-tuned)** [4], trained under identical settings.

For the transformer baselines we replace the language modeling head with a two-layer MLP regression head (256 hidden units, dropout 0.1) and take the [CLS] token as the pooled representation.

4.3 Evaluation Protocol

All models are evaluated with 5-fold cross-validation. For each fold we report Pearson correlation r between predicted and true values, Spearman rank correlation ρ , and root mean squared error (RMSE).

4.4 Implementation Details

Feature extraction was performed with RDKit [12]. Regression was implemented with LightGBM [9]. Transformer fine-tuning was performed with HuggingFace `transformers` [22] on a single NVIDIA T4 GPU. All experiments used a fixed random seed (42) for reproducibility.

5 Results

5.1 Main Comparison on c-Abl

Table 2 summarizes the main result. On the c-Abl dataset, Drugon achieves a Pearson $r = 0.749$, outperforming both fine-tuned transformers: ChemBERTa reaches $r = 0.605$ and MoLFormer $r = 0.551$. The gap is substantial: Drugon improves over ChemBERTa by 24% and over MoLFormer by 36% in terms of explained variance.

Table 2: Main comparison on c-Abl (CHEMBL4550, $n = 1378$). Metrics computed over 5-fold out-of-fold predictions.

Method	r	ρ	RMSE
Binary Morgan + LightGBM	0.723	0.729	0.639
Drugon (Phase Morgan)	0.749	0.753	0.613
ChemBERTa (fine-tuned)	0.605	0.623	0.859
MoLFormer (fine-tuned)	0.551	0.553	0.906

5.2 Compute Comparison

Table 3 contrasts the computational footprint of the three models. Drugon requires approximately $15,000\times$ fewer parameters and $40\times$ less training time than the transformer baselines. Its total model size is 5 MB, compared with $\sim 180\text{--}200$ MB for the transformers.

Table 3: Compute comparison on c-Abl.

Model	Params	Size	Training	r
Drugon	$\sim 3\text{K}$	5 MB	30 s	0.749
ChemBERTa	46 M	180 MB	15 min	0.605
MoLFormer	47 M	200 MB	20 min	0.551

5.3 Cross-Target Evaluation

To test whether the phase-encoding benefit generalizes, we evaluated Drugon on two additional targets (Table 4). In every case, phase encoding provided a consistent, non-negative improvement over the binary Morgan baseline.

Table 4: Cross-target summary. Binary = binary Morgan + LightGBM. Phase = Drugon. Metrics over 5-fold out-of-fold predictions.

Target	ChEMBL ID	n	Binary r	Phase r (Δ)
c-Abl	CHEMBL4550	1378	0.723	0.749 (+0.026)
AChE	CHEMBL220	2000	0.834	0.844 (+0.010)
COX-2	CHEMBL230	2000	0.741	0.742 (+0.000)

The largest gain is observed on c-Abl, where the binding pocket contains a well-defined network of polar contacts that phase-encoding helps distinguish. On COX-2, a hydrophobic pocket, phase encoding provides no benefit — consistent with its interpretation as a signal of chemical polarity.

5.4 Training Stability

To assess the robustness of the result, we re-ran Drugon with five different random seeds for the fold split. The mean Pearson r was 0.746 ± 0.008 , confirming that the reported improvement over binary Morgan is not an artifact of a single split.

6 Discussion

6.1 Why Does a Compact Model Win?

Our results contradict the prevailing trend toward ever-larger pretrained models in molecular machine learning. We attribute the observed advantage of Drugon to three factors.

Data scale mismatch. The transformer baselines contain 46–47 million parameters. For efficient fine-tuning, such models require training sets on the order of hundreds of thousands to millions of examples [23]. Our training sets contain 1378–2000 molecules. The parameter-to-data ratio is therefore on the order of 3×10^4 , far outside the regime in which transformers are known to perform well.

In contrast, Drugon has approximately 3,000 effective parameters, giving a parameter-to-data ratio of order 1.

Architectural alignment. ChemBERTa and MoLFormer are designed for *language modeling* of SMILES strings. Their pretraining objective (masked token prediction) optimizes for the ability to reconstruct local sequence context, which is only indirectly related to binding affinity.

Chemical information density. A single Morgan bit encodes the presence of a specific circular substructure. Phase encoding adds a second dimension of information — *which kind* of chemical context — at negligible cost.

6.2 Interpretation of Phase Encoding

Phase encoding is best understood as a *soft typing* mechanism. Two substructures that hash to the same bit will be distinguished by their accumulated phase. The fact that phase encoding provides the largest benefit on c-Abl (+0.026) and the smallest on COX-2 (+0.000) is informative. The c-Abl ATP-binding pocket contains numerous polar contacts, while COX-2 contains a predominantly hydrophobic channel. This suggests that phase encoding is most useful when binding depends on *chemical specificity* rather than pure shape complementarity.

6.3 Limitations

We highlight three limitations.

1. **Dataset size.** Our conclusions are drawn from three ChEMBL targets with n between 1378 and 2000.
2. **Target diversity.** All three targets are human proteins.
3. **Hyperparameter tuning.** We deliberately used fixed hyperparameters for all fingerprint variants.

6.4 Implications for Practice

For practitioners in the low-data regime, our results suggest the following pragmatic guideline:

When $n < 2000$, prefer compact fingerprint-based models over pretrained transformers.

Drugon trains in 30 seconds on a commodity CPU, requires no GPU, produces interpretable feature importances, and produces a model file small enough to ship inside a web application or mobile app.

7 Conclusion

We introduced Drugon, a compact and efficient method for target-specific binding affinity prediction. Drugon extends the classical Morgan fingerprint with a phase-encoding scheme in which each substructure bit accumulates a complex phase derived from its local chemical environment. Combined with a LightGBM regressor, this representation achieves state-of-the-art accuracy in the low-data regime, outperforming fine-tuned ChemBERTa and MoLFormer on the c-Abl target while requiring $15,000\times$ fewer parameters and $40\times$ less training time.

Across three independent ChEMBL targets, phase encoding provided a consistent, non-negative improvement over the binary Morgan baseline, with the largest gain on the target whose binding pocket depends most on polar interactions.

Our results contribute to a growing body of evidence that, in data-scarce regimes, thoughtful feature engineering remains a competitive strategy in molecular machine learning.

Future work. Several directions merit further study: (i) extension to larger datasets to identify the crossover point at which transformers become advantageous; (ii) phase-encoding of alternative fingerprints; (iii) integration with 3D-aware methods; and (iv) evaluation on prospective, wet-lab-confirmed hits.

Code and data availability. An open-source implementation of Drugon will be released at <https://github.com/mablue/article.drugon> under the MIT license.

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